

Input-oriented FDH Slack Analysis: Formulation and Aggregate Decomposition

slackshare package

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Abstract

We formalize the input-oriented Free Disposal Hull (FDH) slack analysis used by the **slackshare** package. We define FDH-efficient input levels, compute avoidable slack per decision-making unit (DMU), and decompose total input across all DMUs into a system-wide “slack share” and per-DMU contributions. We then extend this analysis to panel data (multiple dates and input types) by analyzing independent cross-sectional slices. The panel extension gracefully handles incomplete data: missing inputs or outputs for individual DMUs are treated as ineligible for analysis at that slice, with results propagated as NA, while aggregate metrics are computed over the observed (eligible) subset using skip-NA semantics. A worked example illustrates the method on synthetic data.

1 Setup and Notation

Consider a set of n decision-making units (DMUs), indexed $i = 1, \dots, n$. Each DMU i consumes a single (typically undesirable) input $x_i \geq 0$ and produces a vector of K outputs $y_i = (y_i^1, \dots, y_i^K) \in \mathbb{R}_{\geq 0}^K$. Throughout, we write vector inequalities component-wise: $y_j \geq y_i$ means $y_j^k \geq y_i^k$ for all $k = 1, \dots, K$.

2 FDH Dominance and Efficient Input

Definition 1 (Dominance set). The dominance set (or dominating set) of DMU i is

$$D(i) = \{j \in \{1, \dots, n\} : y_j \geq y_i\}.$$

By reflexivity, $i \in D(i)$ for all i .

The efficient input level for DMU i is the minimum input among all dominating units:

$$x_i^* = \min_{j \in D(i)} x_j. \tag{1}$$

Since $i \in D(i)$, we have $x_i^* \leq x_i$ always, and x_i^* is well-defined and non-negative (even when $D(i) = \{i\}$ alone).

Proposition 1. *For all i , $x_i^* \leq x_i$. Hence, the slack $slack_i := x_i - x_i^* \geq 0$ is well-defined and non-negative.*

Proof. Since $i \in D(i)$, we have $x_i^* = \min_{j \in D(i)} x_j \leq x_i$. □ □

3 Slack and Efficiency

The slack $\text{slack}_i = x_i - x_i^*$ represents the amount by which DMU i could reduce its input while maintaining at least the same output performance (by replicating any unit in $D(i)$).

A DMU is *FDH-efficient* if and only if $\text{slack}_i = 0$, i.e., $x_i = x_i^*$. Equivalently, a DMU is efficient if it achieves the minimum input among all units that dominate it on every output.

Remark 1. FDH (Free Disposal Hull) imposes only free disposability and allows no convexity. Thus, each DMU is only compared against *actually observed* dominating units, never against hypothetical convex blends. This contrasts with DEA (Data Envelopment Analysis), which constructs a convex frontier and can benchmark against infeasible synthetic units. For a given dataset, FDH yields a coarser (higher) frontier than DEA.

4 Aggregate Decomposition

Define the total input and total slack as

$$X = \sum_{i=1}^n x_i, \quad (2)$$

$$S = \sum_{i=1}^n \text{slack}_i. \quad (3)$$

The *aggregate slack share* measures the fraction of total input across all DMUs that is avoidable excess:

$$\text{slack_share} = \frac{S}{X}. \quad (4)$$

Per-DMU contributions to the system are captured by two share metrics:

$$\text{share_of_total_input}_i = \frac{\text{slack}_i}{X}, \quad (5)$$

$$\text{share_of_total_slack}_i = \begin{cases} \frac{\text{slack}_i}{S} & \text{if } S > 0, \\ 0 & \text{if } S = 0. \end{cases} \quad (6)$$

The first metric expresses each DMU's slack as a fraction of the population's total input. The second expresses each DMU's waste as a fraction of system-wide waste (when any exists).

Proposition 2. When $S > 0$, $\sum_{i=1}^n \text{share_of_total_slack}_i = 1$.

Proof.

$$\sum_{i=1}^n \text{share_of_total_slack}_i = \sum_{i=1}^n \frac{\text{slack}_i}{S} = \frac{1}{S} \sum_{i=1}^n \text{slack}_i = \frac{S}{S} = 1. \quad \square$$

□

5 Validity Conditions

The analysis is well-defined and yields meaningful results under the following preconditions:

1. Each $x_i \geq 0$. (Strictly, $x_i > 0$ avoids division-by-zero in aggregate formulas; $x_i = 0$ requires case-by-case treatment.)

2. For each i , the output vector y_i is not identically zero: $\exists k y_i^k > 0$. (Otherwise, DMU i dominates no other units, and the dominance set $D(i) = \{i\}$ alone, which is valid but offers no insight.)
3. The dataset is non-empty: $n \geq 1$.

6 Worked Numerical Example

Consider a toy dataset with $n = 4$ DMUs, a single undesirable input (e.g., emissions), and one output (e.g., revenue):

DMU	Input x_i	Output y_i	Efficiency
A	100	10	?
B	80	10	?
C	120	15	?
D	60	8	?

Compute the dominance set for each DMU:

- $D(A) = \{j : y_j \geq 10\} = \{A, B, C\}$ (A, B, and C all produce ≥ 10). Thus $x_A^* = \min(100, 80, 120) = 80$.
- $D(B) = \{j : y_j \geq 10\} = \{A, B, C\}$. Thus $x_B^* = \min(100, 80, 120) = 80$.
- $D(C) = \{j : y_j \geq 15\} = \{C\}$ (only C produces ≥ 15). Thus $x_C^* = 120$.
- $D(D) = \{j : y_j \geq 8\} = \{A, B, C, D\}$ (all produce ≥ 8). Thus $x_D^* = \min(100, 80, 120, 60) = 60$.

Compute slack and efficiency:

DMU	x_i	x_i^*	slack $_i$	Efficient?
A	100	80	20	No
B	80	80	0	Yes
C	120	120	0	Yes
D	60	60	0	Yes
Total	360	340	20	—

Here, DMUs B, C, and D are efficient; only A is inefficient with slack of 20 units. The system-wide slack share is $S/X = 20/360 = 0.0556$ (about 5.6% of total input is avoidable).

7 Panel Data Analysis

When data are panel-structured (multiple dates and/or input types), the analysis extends naturally: we analyze independent cross-sectional slices, applying the single-input FDH method to each $(date, input_type)$ pair.

7.1 Balanced Panel (Current Implementation)

For each date $t = 1, \dots, T$ and input type $k = 1, \dots, K_{\text{in}}$, let $x_i^{(t,k)}$ denote the input value for DMU i at date t for input type k , and let $y_i^{(t)}$ denote the (shared) output vector for DMU i at date t (outputs are typically not indexed by input type; the choice of input type does not change the set of achievable outputs in a given date).

In the balanced case, every $(dmu, date, input_type)$ and $(dmu, date, output_type)$ combination is required to be observed. For the slice (t, k) , define the dominance set as

$$D_{t,k}(i) = \{j : y_j^{(t)} \geq y_i^{(t)}\},$$

and the efficient input as

$$x_{i,t,k}^* = \min_{j \in D_{t,k}(i)} x_j^{(t,k)}.$$

The slack, efficiency flag, and aggregate metrics follow as in §6. Crucially, analysis of slice (t, k) is *independent* of all other slices: there is no pooling across dates or input types in the FDH frontier.

7.2 Unbalanced Panel (Implemented)

In practice, panel data are often incomplete: a DMU may not report a given input type, may have begun or ceased operations, or may have missing output values for structural reasons. We now formalize the unbalanced case, in which inputs and outputs may legitimately be absent (NA) rather than present.

Let inputs and outputs take values in $\mathbb{R}_{\geq 0} \cup \{\text{NA}\}$. For each slice (t, k) , define the *eligible* (or observable) set as

$$E_{t,k} = \{i : x_i^{(t,k)} \neq \text{NA} \text{ and } y_i^{(t),j} \neq \text{NA} \text{ for all } j = 1, \dots, K_{\text{out}}\}, \quad (7)$$

i.e., the set of DMUs with fully observed input and fully observed output vector at slice (t, k) .

Only DMUs in $E_{t,k}$ can participate in the analysis: dominance cannot be assessed without knowing all output dimensions, and an unobserved input cannot be minimized. Redefine the dominance set as

$$D_{t,k}(i) = \{j \in E_{t,k} : y_j^{(t)} \geq y_i^{(t)}\}, \quad \text{for } i \in E_{t,k}.$$

The efficient input is then

$$x_{i,t,k}^* = \begin{cases} \min_{j \in D_{t,k}(i)} x_j^{(t,k)} & \text{if } i \in E_{t,k}, \\ \text{NA} & \text{if } i \notin E_{t,k}. \end{cases} \quad (8)$$

Slack and efficiency are likewise NA for any DMU outside $E_{t,k}$:

$$\text{slack}_{i,t,k} = \begin{cases} x_i^{(t,k)} - x_{i,t,k}^* & \text{if } i \in E_{t,k}, \\ \text{NA} & \text{if } i \notin E_{t,k}. \end{cases} \quad (9)$$

Remark 2. Note that Proposition 1 still holds on the eligible set: for $i \in E_{t,k}$, self-membership $i \in D_{t,k}(i)$ ensures $x_{i,t,k}^* \leq x_i^{(t,k)}$, hence $\text{slack}_{i,t,k} \geq 0$.

Aggregate metrics are computed over the observable sub-population only (analogous to `pandas skipna=True` semantics):

$$X_{t,k} = \sum_{i \in E_{t,k}} x_i^{(t,k)}, \quad (10)$$

$$S_{t,k} = \sum_{i \in E_{t,k}} \text{slack}_{i,t,k}, \quad (11)$$

$$\text{slack_share}_{t,k} = \begin{cases} \frac{S_{t,k}}{X_{t,k}} & \text{if } E_{t,k} \neq \emptyset, \\ \text{NA} & \text{if } E_{t,k} = \emptyset. \end{cases} \quad (12)$$

Note that $X_{t,k}$ and $S_{t,k}$ are always computed sums; when $E_{t,k} = \emptyset$, these are zero (empty sum), and only the ratio $\text{slack_share}_{t,k}$ is NA. A single incomplete DMU does not invalidate the slice's statistics; aggregates remain valid over the eligible subset.

Proposition 3 (Balanced case as special case). *The balanced case ($E_{t,k} = \{1, \dots, n\}$ for all (t, k)) is recovered as a special instance of the unbalanced formulation.*

Proof. When all inputs and outputs are observed, $E_{t,k} = \{1, \dots, n\}$, and the formulas for $D_{t,k}(i)$, $x_{i,t,k}^*$, $\text{slack}_{i,t,k}$, and aggregate metrics reduce identically to the balanced definitions. $\square$ $\square$

7.2.1 Micro-Example: Unbalanced Data

Continuing the toy dataset from §6, suppose we analyze this data at a specific date and input type, but DMU C's output is not recorded (NA) at this time. The observable set is $E = \{A, B, D\}$, and C receives an entirely NA row.

Dominance sets restricted to E :

- $D(A) = \{j \in E : y_j \geq 10\} = \{A, B\}$. Thus $x_A^* = \min(100, 80) = 80$.
- $D(B) = \{j \in E : y_j \geq 10\} = \{A, B\}$. Thus $x_B^* = \min(100, 80) = 80$.
- $D(C) = \text{NA}$ (C is not in E).
- $D(D) = \{j \in E : y_j \geq 8\} = \{A, B, D\}$. Thus $x_D^* = \min(100, 80, 60) = 60$.

Results:

DMU	x_i	y_i	x_i^*	Slack
A	100	10	80	20
B	80	10	80	0
C	120	NA	NA	NA
D	60	8	60	0
Total (E only)	240	—	220	20

The slice's aggregate metrics sum only over $E = \{A, B, D\}$: $X = 240$, $S = 20$, $\text{slack_share} = 20/240 = 0.0833$ (8.33%).

7.2.2 Practical Motivation

Missing values arise naturally in panel settings: a DMU may not yet be reporting a given input type, may have ceased operations at a certain date, or may simply lack data collection for a metric. The unbalanced formulation treats such gaps not as errors but as structural features of the data, preserving the available information for analysis (the complete-data DMUs) while flagging the incomplete rows.

Remark 3. The unbalanced extension is implemented in the current `analyze_panel` function (see `slackshare/panel.py`), which gracefully handles missing/unbalanced data by NA propagation and eligible-set aggregation, exactly as formalized in this section. Prior versions enforced complete data; the current version allows missing entries per (dmu, date, input/output) combination and treats them as ineligible, matching the mathematical framework above.

8 Discussion and Limitations

The FDH approach offers several advantages and limitations:

- **No hypothetical units.** Unlike DEA, FDH never compares against infeasible synthetic units, reducing the risk of spurious efficient targets.
- **Single-input restriction.** The core analysis considers only one input at a time. Multi-input problems must be solved separately per input (as in `slackshare`), and the results do not form a unified multi-input efficiency measure.
- **Coarse frontier.** FDH’s lack of convexity yields a coarser frontier than DEA, potentially missing efficiency gains from convex combinations (e.g., merging or scaling DMUs). For some applications, this is a feature (only observed feasible points matter); for others, a limitation.
- **Sample size sensitivity.** When n is small, FDH frontiers can be degenerate (e.g., every DMU may be efficient if outputs are sparse or multi-dimensional). Larger datasets typically yield more informative slack patterns.